The Origin of CP Violation from Recognition Science: CPT Invariance, Ledger Phase Structure, and the 8-Tick Epoch

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

March 2026

Abstract

We derive the origin of CP violation from the Recognition Science (RS) ledger structure [4]. CPT invariance is automatic: the combined operation of charge conjugation (C: charge $\to -$charge), parity (P: position $\to -$position), and time reversal (T: tick $\to (7 - \text{tick}) \bmod 8$) preserves the cost functional $J(x)$ and ledger balance. CPT is involutive ($\text{CPT}^2 = \text{id}$). CP violation arises because C and P individually preserve cost but their product CP does not preserve the 8-tick phase ordering: the forward and backward phase sequences are inequivalent when weak-sector entries carry complex phases from the three-generation CKM matrix. Since CPT is exact, CP violation is the complement of T violation. The CPT mass bound $|\Delta m/m| < 10^{-9}$ is satisfied automatically. We prove that CP violation requires $N_{\text{gen}} \geq 3$ (two generations give no CP-odd phase), connecting this result to the RS prediction $N_{\text{gen}} = 3$. The *existence* of CP violation is a structural consequence; the *magnitude* $\mathcal{J} \approx 3.2 \times 10^{-5}$ is an open calculation.

1 Introduction

CP violation—matter–antimatter asymmetry in weak interactions—was discovered in 1964 in neutral kaon decays [1]. The Standard Model accommodates CP violation through the Kobayashi–Maskawa mechanism: with $N_{\text{gen}} \geq 3$, the quark mixing matrix carries an irreducible complex phase [2].

However, the Standard Model does not explain:

- Why CP is violated (why the phase is non-zero).
- Why CPT is preserved (the CPT theorem relies on Lorentz invariance, locality, and unitarity—assumptions about the continuum).
- Why there are exactly three generations.

RS answers all three from the ledger structure.

2 Recognition Science Framework

RS derives all physical law from three axioms [4]:

(A1) $J(1) = 0$ *(Normalization)*

(A2) $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$ *(Recognition Composition Law)*

(A3) $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$ *(Calibration)*

Theorem 2.1 (Cost Uniqueness [4]). *The unique continuous solution is $J(x) = \frac{1}{2}(x + x^{-1}) - 1$, satisfying $J(x) \geq 0$ (equality iff $x = 1$) and $J(x) = J(x^{-1})$.*

Proof sketch. $H(t) := 1 + J(e^t)$ satisfies the d'Alembert equation $H(s+t) + H(s-t) = 2H(s)H(t)$ with $H(0) = 1$, $H''(0) = 1$. By the Aczél classification, $H(t) = \cosh t$, giving $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. □

Each ledger entry e carries a positive ratio $x_e > 0$ (the recognition scale ratio between debit and credit). The discrete symmetries act on these ratios and on the phase structure of the 8-tick epoch.

3 Discrete Symmetries on the Ledger

Definition 3.1 (Charge Conjugation C). On the recognition ledger, C maps charge $q \to -q$ (equivalently, it swaps debit and credit entries). Since $J(x)$ depends on $|q|$ through the gap weight, J is invariant: $J(C(e)) = J(e)$.

Definition 3.2 (Parity P). P maps spatial position $\mathbf{x} \to -\mathbf{x}$. The cost functional depends on $|\mathbf{x}|$ (distance), not direction, so $J(P(e)) = J(e)$.

Definition 3.3 (Time Reversal T). On the 8-tick epoch, T maps tick $t \to (7 - t) \bmod 8$. This reverses the phase sequence within the epoch. T preserves cost: $J(T(e)) = J(e)$.

4 CPT Invariance

Theorem 4.1 (CPT Preservation). *CPT preserves cost:*

$$J(\mathrm{CPT}(e)) = J(e) \tag{1}$$

for every ledger entry e. CPT also preserves ledger balance (total charge conservation).

Proof. We verify cost-preservation for each transformation individually.

Charge conjugation C: C negates the charge $q \to -q$, which at the ratio level maps $x_e \to x_e^{-1}$ (debits become credits and vice versa). By Theorem 2.1, $J(x^{-1}) = J(x)$, so $J(C(e)) = J(x_e^{-1}) = J(x_e) = J(e)$.

Parity P: P maps $\mathbf{x} \to -\mathbf{x}$. The cost functional $J(x_e)$ depends on the scale ratio $x_e > 0$, not on the spatial position. Hence $J(P(e)) = J(e)$.

Time reversal T: T maps tick $t \to (7 - t) \bmod 8$, reversing the epoch phase ordering. $J(x_e)$ depends on x_e, not on the tick index. Hence $J(T(e)) = J(e)$.

Composition: since all three preserve J, so does their product: $J(\mathrm{CPT}(e)) = J(e)$.

Ledger balance is preserved because C swaps debits and credits (preserving the total), P and T do not alter charge values. □

Theorem 4.2 (CPT Involutive)**.** $\mathrm{CPT}^2 = \mathrm{id}$.

Proof. $C^2 = \mathrm{id}$ (charge double-negation: $q \to -q \to q$). $P^2 = \mathrm{id}$ (position double-negation: $\mathbf{x} \to -\mathbf{x} \to \mathbf{x}$). $T^2 = \mathrm{id}$ (tick reversal: $t \to (7-t) \to (7-(7-t)) = t \bmod 8$). Since C, P, T act on independent degrees of freedom (charge, position, and tick, respectively), they commute: $CPT = CTP = PCT = \ldots$. Therefore $(\mathrm{CPT})^2 = C^2P^2T^2 = \mathrm{id}$. □

Corollary 4.3 (CPT Consequences)**.** *CPT invariance implies:*

(i) $m_{\text{particle}} = m_{\text{antiparticle}}$.

(ii) $\tau_{\text{particle}} = \tau_{\text{antiparticle}}$ *(equal lifetimes).*

(iii) $q_{\text{particle}} = -q_{\text{antiparticle}}$ *(opposite charges).*

5 Why CP Is Violated

Proposition 5.1 (CP Violation from Phase Structure)**.** *When weak-sector ledger entries carry complex CKM phases (imported from the Standard Model), CP does not preserve the 8-tick phase ordering: the forward phase sequence and the CP-transformed sequence are inequivalent whenever $N_{\text{gen}} \geq 3$.*

Proof. The 8-tick epoch has a natural forward direction: ticks proceed $0, 1, 2, \ldots, 7$ with phases $e^{2\pi ik/8}$ for $k = 0, \ldots, 7$. Under CP, the charge and spatial parity are reversed, but the tick ordering is preserved. The weak-sector entries carry generation-dependent phases from the CKM matrix: $V_{ij} = |V_{ij}|e^{i\delta_{ij}}$. Under CP: $V_{ij} \to V_{ij}^*$ (complex conjugation). If $\mathrm{Im}(V_{ij}) \neq 0$ (which requires $N_{\text{gen}} \geq 3$), then $V_{ij} \neq V_{ij}^*$, and the forward phase sequence differs from the CP-transformed sequence. This phase mismatch is CP violation. □

Proposition 5.2 (CP Requires Three Generations)**.** *CP violation vanishes for $N_{\text{gen}} < 3$.*

Proof. For $N_{\text{gen}} = 2$, the 2×2 CKM matrix has one real parameter (the Cabibbo angle θ_C) and no complex phase. $V_{ij} = V_{ij}^*$ for all i, j, so CP is preserved.

For $N_{\text{gen}} = 3$, the 3×3 CKM matrix has three angles and one irreducible phase δ, giving $\mathcal{J} = \mathrm{Im}(V_{us}V_{cb}V_{ub}^*V_{cs}^*) \neq 0$. (This is the Kobayashi–Maskawa theorem [2].) □

Remark 5.3 (Status of the RS derivation of CP violation). RS forces $N_{\text{gen}} = 3$ from the face-pair count of the $D = 3$ cube (three pairs of opposite faces). Since the minimum number of generations for a non-trivial CKM phase is $N_{\text{gen}} = 3$ (Kobayashi–Maskawa theorem), CP violation is a *structural consequence* of the RS dimension-forcing.

However, this argument has two levels:

(i) RS derives $N_{\text{gen}} = 3$ (via $D = 3$ cube geometry).

(ii) The KM theorem (standard group theory) then implies that a 3×3 unitary matrix generically has one irreducible complex phase.

Step (i) is the RS content; step (ii) is imported from the Standard Model. The RS framework predicts the *existence* of CP violation (because 3 generations are forced and 3 is the minimum for KM phases), but does not independently derive the *magnitude* of the Jarlskog invariant $\mathcal{J} \approx 3.2 \times 10^{-5}$. Predicting $\mathcal{J}$ from the RS CKM matrix structure (rungs, gap weights, and three-generation ladder structure) is an identified open problem.

6 T Violation as the Complement

Corollary 6.1 (CP Violation $\Leftrightarrow$ T Violation)**.** *Since CPT is exact (Theorem 4.1), CP violation implies T violation, and conversely.*

Proof. If CPT is a symmetry but CP is not, then $T = (CP)^{-1}(CPT)$ cannot be a symmetry either (the composition of a non-symmetry with a symmetry is not a symmetry). Conversely, if T is a symmetry and CPT is a symmetry, then $CP = (CPT) \cdot T^{-1}$ would also be a symmetry, contradicting CP violation. Hence CP non-invariance $\Leftrightarrow$ T non-invariance. □

Remark 6.2. The CPLEAR experiment [3] and the BaBar/Belle B-factories have directly observed T violation in the neutral kaon and B-meson systems, confirming this complementarity.

7 CPT Tests

Test	**Experimental bound**	**RS prediction**
$\lvert m_p - m_{\bar{p}}\rvert/m_p$	$< 7 \times 10^{-10}$ [5]	0 (exact)
$\lvert q_p + q_{\bar{p}}\rvert/e$	$< 7 \times 10^{-10}$ [5]	0 (exact)
$\lvert\tau_{\mu^+} - \tau_{\mu^-}\rvert/\tau$	$< 2 \times 10^{-5}$ [5]	0 (exact)
$\lvert m_{K^0} - m_{\bar{K}^0}\rvert/m_{K^0}$	$< 6 \times 10^{-19}$ [5]	0 (exact)

8 Predictions

1. CPT **exact**: All CPT tests should yield null results. Any CPT violation would falsify RS.

2. CP **violation requires 3 generations**: Confirmed experimentally.

3. T **violation observed**: Confirmed by CPLEAR and B-factories.

4. **No CPT-violating Lorentz violation**: SME (Standard Model Extension) coefficients for CPT violation are all zero.

9 Conclusion

CPT invariance is automatic on the recognition ledger, following from the individual cost-preserving nature of C, P, and T. CP violation arises because the 8-tick phase structure is not preserved under CP when weak-sector entries carry complex CKM phases, which require $N_{\text{gen}} \geq 3$. Since RS forces $N_{\text{gen}} = 3$, CP violation is a structural consequence: three generations are necessary and sufficient for the KM mechanism to produce an irreducible phase (see Remark 5.3).

References

[1] J. H. Christenson, J. W. Cronin, V. L. Fitch, and R. Turlay, "Evidence for the 2π Decay of the K_2^0 Meson," Phys. Rev. Lett. **13**, 138 (1964).

[2] M. Kobayashi and T. Maskawa, "CP-Violation in the Renormalizable Theory of Weak Interaction," Prog. Theor. Phys. **49**, 652 (1973).

[3] CPLEAR Collaboration, "First Direct Observation of Time-Reversal Non-Invariance in the Neutral-Kaon System," Phys. Lett. B **444**, 43 (1998).

[4] J. Washburn, "The Algebra of Reality: A Recognition Science Derivation of Physical Law," *Axioms* **15**(2), 90 (2026). doi:10.3390/axioms15020090

[5] S. Navas *et al.* (Particle Data Group), "Review of Particle Physics," Phys. Rev. D **110**, 030001 (2024).